

Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

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Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

Part 4: Configure OSPF Passive Interfaces

- g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/0

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? 129

Does R2 show up as an OSPF neighbor on R1? YES

Does R2 show up as an OSPF neighbor on R3? NO

What does this information tell you?

It indicates that configuring an interface as passive prohibits OSPF from sending out Hello packets, so no neighbouring relationship can exist between R2 and R3. Nevertheless, R2 does advertise its route to the rest of the network because it does have an active connection to R1

- h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

Enter global configuration mode, enter the OSPF process with router ospf 1, and apply the no passive-interfaces 0/0/1 command.

- i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? Serial0/0/1

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

The accumulated cost is 65. This is determined by adding the default cost of R3's active Serial0/0/1 interface (64) to the cost of R2's GigabitEthernet0/0 interface (1)

Is R2 listed as an OSPF neighbor to R3? YES

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

- a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
```

```
R1(config-router)# auto-cost reference-bandwidth 100
```

```
% OSPF: Reference bandwidth is changed.
```

```
Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

OSPF's reference bandwidth is 100 Mbps, which means links at 100 Mbps or above will have a cost of 1. By modifying the reference bandwidth, OSPF can accurately measure the cost of Gigabit or 10-Gigabit links so that it can determine which is the actual fastest path.

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

For the 192.168.3.0/24 network, the total cost is 782. This combines R1's modified S0/0/1 cost (781) with R3's G0/0 cost (1).

- i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

The new cost is 1562. This happens because the bandwidth 128 command makes OSPF treat the serial links as 128 kbps, resulting in a much higher cost of 781 per link. Since the route crosses two of these serial links to reach the destination, the total cost adds up to 1562 (781 + 781).

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

By manually assigning an exorbitant price for R1's dedicated serial link (cost of 1565), this became a better path option than the serial link through R2 (total cost 1563). Therefore, OSPF has chosen to keep the more suitable path in the routing table

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

If a Router ID isn't manually configured (or tied to a stable loopback address), OSPF relies on the highest active physical IP address. If that physical interface fails, the Router ID changes, which tears down all OSPF adjacencies and forces the entire network to undergo an SPF recalculation

2. Why is the DR/BDR election process not a concern in this lab?

Elections for both a DR and BDR are used on multi-access broadcast networks. However, since this lab uses only point-to-point serial links, there is no need to complete DR/BDR elections

3. Why would you want to set an OSPF interface to passive?

To reduce unnecessary OSPF processing and causing congested network traffic, improving the performance of the network

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

By dividing the network into multiple areas, OSPF limits routing updates to local zones, resulting in smaller routing tables and high scalability

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

'network 10.0.0.0 0.255.255.255 area 0'

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

The EIGRP route will be selected. When comparing different routing protocols, routers ignore metrics and rely on AD. EIGRP has a lower AD (90) than OSPF (110) and RIP (120).